



DAYANANDA SAGAR
UNIVERSITY

DAYANANDA SAGAR UNIVERSITY
SCHOOL OF ENGINEERING
DEPARTMENT OF CSE (DATA SCIENCE)



**SCHOOL OF
ENGINEERING**



PRESENTED BY LOGICFUSION

Compound Emotion Recognition System

UNDER THE SUPERVISION

PROF. SHIVAMMA

ASSISTANT PROFESSOR



DAYANANDA SAGAR
UNIVERSITY



SCHOOL OF
ENGINEERING

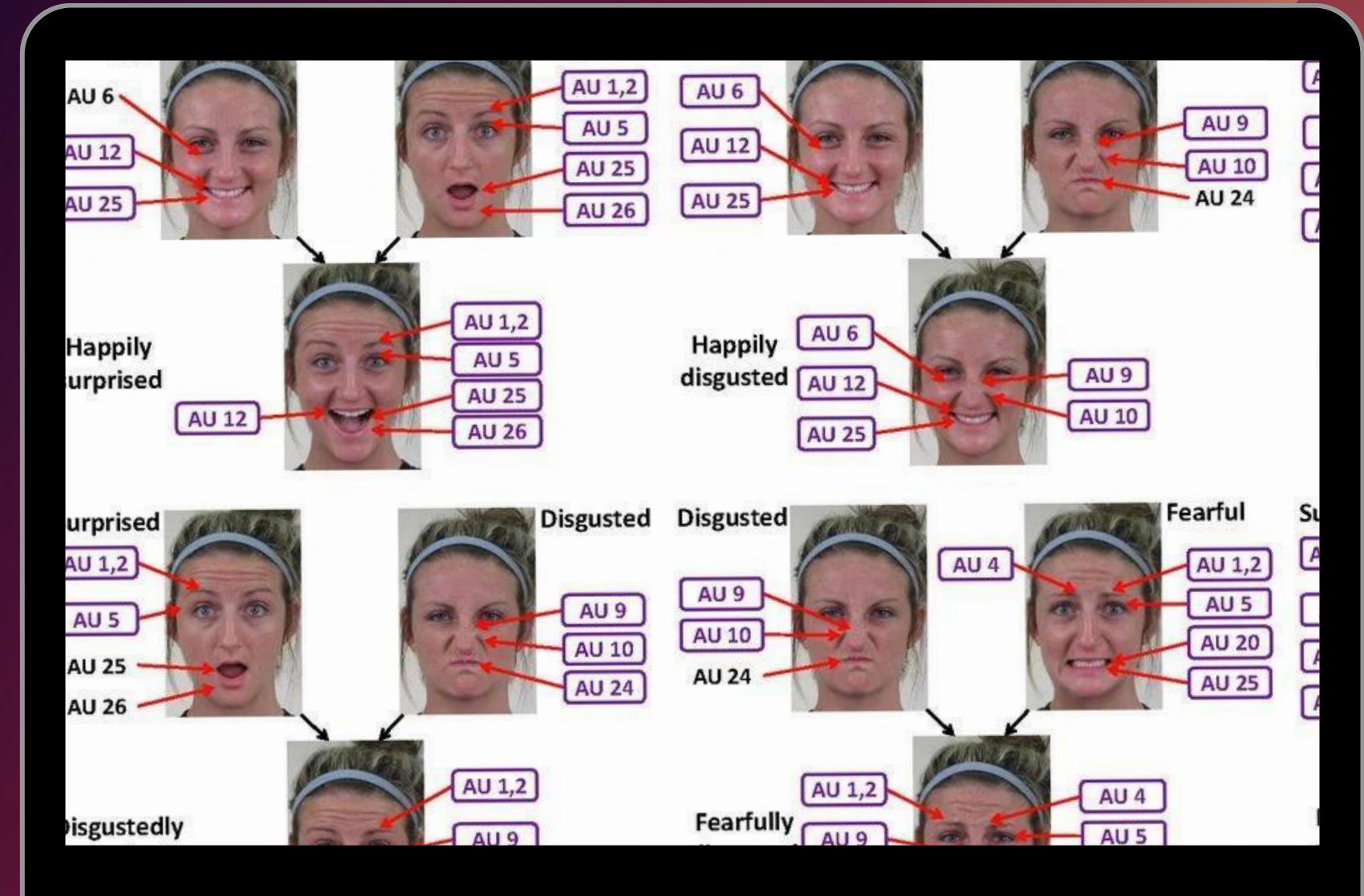


ABSTRACT

Facial expression recognition is a critical component of Human-Computer Interaction (HCI), yet most existing systems are limited to detecting only basic emotions like happiness or anger. This project proposes a robust system for recognizing "Compound Emotions"—complex emotional states such as "Happily Surprised" or "Fearfully Angry"—by analyzing the morphological features of the face. Instead of relying solely on pixel data, our approach extracts geometric landmarks from the face (eyes, eyebrows, mouth) and calculates the Euclidean distances between them to model the precise muscle movements associated with these complex expressions. This method aims to improve detection accuracy in real-world scenarios where human emotions are rarely singular or static.

PROBLEM STATEMENT.

Current Facial Expression Recognition (FER) systems suffer from a "Basic Emotion Bias," where they force-fit complex human expressions into one of six standard categories. This leads to significant misclassification; for instance, a person who is "Happily Surprised" might be incorrectly labeled as just "Happy," ignoring the surprise element entirely. Furthermore, standard pixel-based classifiers are often sensitive to lighting and skin tone variations. The core problem this project addresses is the lack of a geometric, morphology-based framework that can reliably distinguish and classify these overlapping compound emotions with high precision.





INTRODUCTION

Non-verbal communication accounts for a significant portion of human interaction, with facial expressions playing a pivotal role in conveying intent and emotion. While traditional computer vision models have achieved high accuracy in controlled environments, they often fail to capture the nuance of spontaneous, mixed emotions that occur in daily life. This project focuses on bridging that gap by developing a system capable of identifying compound facial expressions. By leveraging morphological analysis, we aim to create a model that understands the structural changes of the face, enabling applications in fields ranging from mental health monitoring to advanced user experience testing.



SOLUTION OVERVIEW.

CORE ARCHITECTURE

- Morphological Engine: Utilizes a 68-point landmark detection system to map facial geometry.
 - Vector Analysis: Calculates Euclidean distances and angles between features (e.g., lip curvature) instead of processing raw pixels.
- Hybrid Classification: Identifies specific overlaps in muscle movements (Action Units) to detect complex states.

REAL-WORLD IMPACT

- Beyond Basic Tags: Detects nuanced human behavior like "Happily Surprised" rather than just "Happy".
- Critical Applications: Enables precise monitoring for mental health diagnostics, driver fatigue systems, and genuine user experience analysis.

KEY INNOVATION

- Geometry Over Pixels: Our distance-based approach is robust against poor lighting and skin tone variations that confuse standard cameras.
- Compound Precision: Uniquely trained to resolve the "Basic Emotion Bias" by modeling the structural collision of two simultaneous emotions.



LITERATURE SURVEY

Our comprehensive review of recent literature confirms that single-feature models are insufficient for capturing the nuance of human expression, driving the industry toward hybrid and geometric approaches. Research into hybrid feature extraction demonstrates that combining geometric landmarks with texture analysis significantly outperforms traditional single-method models. Furthermore, foundational studies on the iCV-MEFED dataset validate the existence of 50 distinct compound emotion categories, providing the necessary structural framework to model complex states like "Fearfully Angry" rather than just basic emotions. Comparative surveys further support our choice of a morphological (landmark-based) architecture, highlighting its superior robustness to environmental factors like lighting variations and skin tone differences compared to standard pixel-based Deep

PAPER -1 : FACIAL EXPRESSION RECOGNITION USING HYBRID FEATURES OF PIXEL AND GEOMETRY

We reference this paper to justify using Geometric Features (distances between landmarks) combined with texture data, as this hybrid approach is proven to outperform single-method models.

PAPER -2 : COMPOUND FACIAL EMOTION RECOGNITION (ICV-MEFED DATASET)

We use this study to define the 50 distinct categories of compound emotions and to understand the specific muscle activation overlaps (e.g., how "Fear" and "Surprise" share specific brow movements).

PAPER -3 : DEEP LEARNING APPROACHES FOR FACIAL EXPRESSION RECOGNITION (BDCC)

We refer to this for our CNN architecture design, specifically how Convolutional Neural Networks can be optimized to extract features from facial images efficiently.

PAPER -4 : SURVEY ON FACIAL EXPRESSION RECOGNITION TECHNIQUES

We use this survey as a baseline comparison to demonstrate why our morphological approach is more robust against lighting changes compared to older techniques like LBP (Local Binary Patterns).



TECHNOLOGY & METHODOLOGY

TECHNOLOGY

- Language: Python (for all core logic and processing).
- Computer Vision: OpenCV & MediaPipe (for real-time face detection and landmark meshing).
- Machine Learning: TensorFlow/Keras or Scikit-Learn (for the classification model).
- Data Processing: NumPy & Pandas (for calculating geometric vectors).



METHODOLOGY

- 1. Face Acquisition:** The system captures video frames and converts them to grayscale to reduce computational load.
- 2. Landmark Localization:** We utilize a pre-trained model (like MediaPipe) to map 68 or 468 specific points (fiducial points) on the user's face.
- 3. Morphological Calculation:** The core innovation involves calculating Euclidean Distances between key pairs (e.g., the distance between the upper and lower lip to measure "openness," or the inner brow distance to measure "frown intensity").
- 4. Classification:** These calculated distance vectors are fed into our classifier to predict the exact compound emotion label.

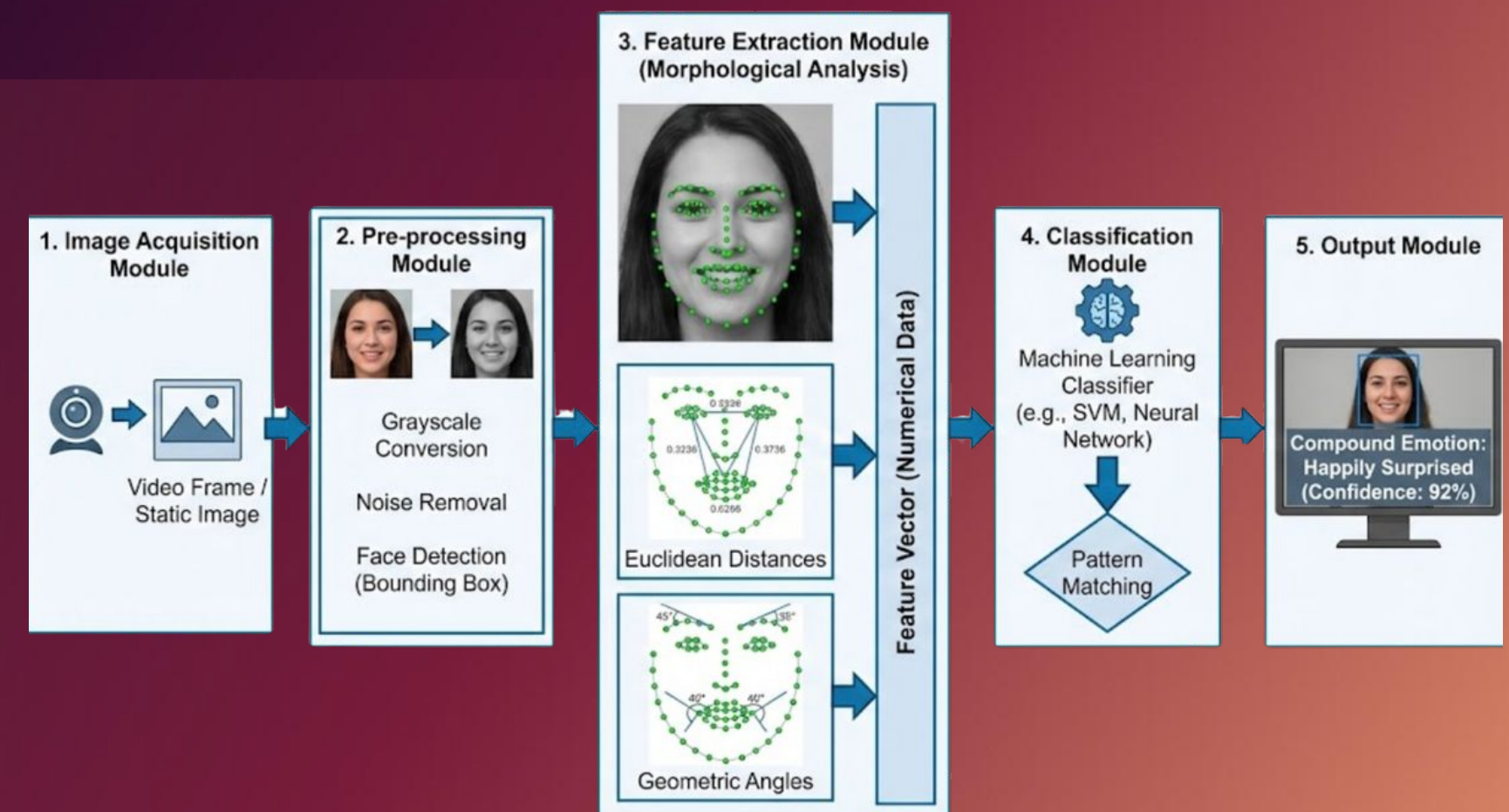




SYSTEM ARCHITECTURE



- **input Module:** Captures raw image data from the camera.
- **Preprocessing:** Normalizes the face size and orientation (alignment) so head tilt doesn't affect calculations.
- **Feature Extraction Layer:** The "Morphological Engine" that turns pixels into mathematical coordinates and distance vectors.
- **Decision Layer:** The trained ML model that outputs the probability of each emotion (e.g., 80% Happy, 20% Surprise).





RESEARCH PAPERS.



Here are the list of research paper we have study:

Paper Title	Authors
CNN-Transformer Architecture Solution for Compound Facial Expression Recognition	Sana Ullah, Yuanlun Xie, Jie Ou, and Wenhong Tian
Compound Facial Expression Recognition Based on Graph Convolutional Multi-Label Learning	Juan Hu, Bin Li, Xiangwei Mou, and Mengchen Yan
Deep Learning Based Analysis for Facial Expression Recognition of Simple and Complex Emotions in Healthcare	L. Monish, D. Shivamma, M. Manjula, and Pooja Shree H. R.
Dominant and Complementary Emotion Recognition From Still Images of Faces	Jianzhu Guo, Zhen Lei, Jun Wan, Egils Avots, et al.
Facial Expression Recognition for Compound Emotions using Mobile Net Architecture	Chang Liu, Kaoru Hirota, Junjie Ma, Zhiyang Jia, and Yaping Dai



Paper Title	Authors
Facial geometric feature extraction based emotional expression classification using machine learning algorithms	Sana Ullah, Yuanlun Xie, Jie Ou, and Wenhong Tian
Deep Learning Approaches for Facial Expression Recognition	Juan Hu, Bin Li, Xiangwei Mou, and Mengchen Yan
Survey on Facial Expression Recognition Techniques	L. Monish, D. Shivamma, M. Manjula, and Pooja Shree H. R.
Facial expression recognition in video with multiple feature fusion	J. Chen, Z. Chen, Z. Chi, and H. Fu
A compact deep learning model for robust facial expression recognition	C.-M. Kuo, S.-H. Lai, and M. Sarkis
Deep pain: Exploiting long shortterm memory networks for facial expression classification	P. Rodriguez, G. Cucurull, J. Gonàlez, J. M. Gonfaus, K. Nasrollahi, T. B. Moeslund, and F. X. Roca
Facial Expression Recognition Using Hybrid Features of Pixel and Geometry	M. Murugappan and A. Mutawa



CONCLUSION

In conclusion, this phase of the project has established a strong theoretical foundation for detecting compound emotions using geometric analysis. We have successfully defined the limitations of existing systems and identified the specific morphological features—such as eye openness and mouth curvature—that are critical for distinguishing complex expressions. The literature survey has confirmed that a hybrid or geometry-focused approach is the most viable path forward. Our next immediate steps are to finalize the dataset generation and begin training the classification model.





DAYANANDA SAGAR
UNIVERSITY



SCHOOL OF
ENGINEERING



THANK YOU